Lesson 1.01 **Duration: 40 minutes**

What is ICT?

Learning Objectives

Timing Breakdown

0-5 min	**Engage**	Ask students: 'Have you ever seen a computer? What did it do?' Write all respons...
5-15 min	**Explore**	Show the parts of a computer on a diagram or real machine. Point to each part an...
15-25 min	**Explain**	Define computer: 'An electronic device that accepts data, processes it, and prod...
25-35 min	**Elaborate**	Present scenario: 'Your school needs to maintain records for 200 students — mark...
35-40 min	**Evaluate**	Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does ...

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 KB, MB, GB — place value and multiplication
Science: Electricity powers all computer components — circuits and conductors
Language: Labeling diagrams builds technical vocabulary and precise writing

Digital Citizen Reminder: Handle computer equipment carefully — it is expensive and shared by all students.

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does. Use pencil and paper only.

Resources Needed:
- Computer diagram or real computer
- labeled chart
- exit ticket paper

- chalkboard

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab — list parts, prices, and total budget
SEND: Touch and identify real computer parts with teacher guidance, simplified labeling

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down a computer system into input, processing, memory, and output components

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.02	**Duration: 40 minutes**

History of Computers

Learning Objectives

- Identify major generations of computers
- Explain how computers have changed over time
- Compare early computers with modern ones

Timing Breakdown

0-5 min	**Engage**	Ask students: 'Have you ever seen a computer? What did it do?' Write all respons...
5-15 min	**Explore**	Show the parts of a computer on a diagram or real machine. Point to each part an...
15-25 min	**Explain**	Define computer: 'An electronic device that accepts data, processes it, and prod...
25-35 min	**Elaborate**	Present scenario: 'Your school needs to maintain records for 200 students — mark...
35-40 min	**Evaluate**	Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does ...

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 KB, MB, GB — place value and multiplication
Science: Electricity powers all computer components — circuits and conductors
Language: Labeling diagrams builds technical vocabulary and precise writing

Digital Citizen Reminder: Handle computer equipment carefully — it is expensive and shared by all students.

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does. Use pencil and paper only.

Resources Needed:
- Computer diagram or real computer
- labeled chart
- exit ticket paper
- chalkboard

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab — list parts, prices, and total budget
SEND: Touch and identify real computer parts with teacher guidance, simplified labeling

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down a computer system into input, processing, memory, and output components

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.03	Duration: 40 minutes

Introduction to Internet

Learning Objectives

- Define the Internet and its basic function
- Explain how devices connect through networks
- Identify services available on the Internet

Timing Breakdown

0-5 min	**Engage**	Ask students: 'Have you ever seen a computer? What did it do?' Write all respons...
5-15 min	**Explore**	Show the parts of a computer on a diagram or real machine. Point to each part an...
15-25 min	**Explain**	Define computer: 'An electronic device that accepts data, processes it, and prod...
25-35 min	**Elaborate**	Present scenario: 'Your school needs to maintain records for 200 students — mark...
35-40 min	**Evaluate**	Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does ...

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 KB, MB, GB — place value and multiplication
Science: Electricity powers all computer components — circuits and conductors
Language: Labeling diagrams builds technical vocabulary and precise writing

Digital Citizen Reminder: Handle computer equipment carefully — it is expensive and shared by all students.

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does. Use pencil and paper only.

Resources Needed:
- Computer diagram or real computer
- labeled chart
- exit ticket paper
- chalkboard

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab — list parts, prices, and total budget
SEND: Touch and identify real computer parts with teacher guidance, simplified labeling

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down a computer system into input, processing, memory, and output components

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.04 | Duration: 40 minutes

Web Browsers

Learning Objectives

- Identify popular web browsers and their icons
- Open a browser and navigate to a website
- Explain the parts of a browser window

Timing Breakdown

0-5 min	**Engage**	Ask students: 'Have you ever seen a computer? What did it do?' Write all respons...
5-15 min	**Explore**	Show the parts of a computer on a diagram or real machine. Point to each part an...
15-25 min	**Explain**	Define computer: 'An electronic device that accepts data, processes it, and prod...
25-35 min	**Elaborate**	Present scenario: 'Your school needs to maintain records for 200 students — mark...
35-40 min	**Evaluate**	Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does ...

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: ‘An electronic device that accepts data, processes it, and produces output.’ Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: ‘Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?’ Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: ‘1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.’ Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 KB, MB, GB — place value and multiplication
Science: Electricity powers all computer components — circuits and conductors
Language: Labeling diagrams builds technical vocabulary and precise writing

Digital Citizen Reminder: Handle computer equipment carefully — it is expensive and shared by all students.

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does. Use pencil and paper only.

Resources Needed:

- Computer diagram or real computer
- labeled chart
- exit ticket paper
- chalkboard

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab — list parts, prices, and total budget
SEND: Touch and identify real computer parts with teacher guidance, simplified labeling

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down a computer system into input, processing, memory, and output components

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.05	Duration: 40 minutes

Search Engines

Learning Objectives

- Define search engines and name popular ones
- Perform a basic search with keywords
- Evaluate search results for relevance

Timing Breakdown

0-5 min	Engage	Ask students: 'Have you ever seen a computer? What did it do?' Write all respons...
5-15 min	Explore	Show the parts of a computer on a diagram or real machine. Point to each part an...
15-25 min	Explain	Define computer: 'An electronic device that accepts data, processes it, and prod...
25-35 min	Elaborate	Present scenario: 'Your school needs to maintain records for 200 students — mark...
35-40 min	Evaluate	Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does ...

Engage (5-7 minutes)

Teacher Activity

Ask students: ‘Have you ever seen a computer? What did it do?’ Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: ‘All these involve processing information — that is what computers do.’

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: ‘What do you think this does?’ Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 KB, MB, GB — place value and multiplication
Science: Electricity powers all computer components — circuits and conductors
Language: Labeling diagrams builds technical vocabulary and precise writing

Digital Citizen Reminder: Handle computer equipment carefully — it is expensive and shared by all students.

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does. Use pencil and paper only.

Resources Needed:

- Computer diagram or real computer
- labeled chart
- exit ticket paper
- chalkboard

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab — list parts, prices, and total budget
SEND: Touch and identify real computer parts with teacher guidance, simplified labeling

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down a computer system into input, processing, memory, and output components

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.06	Duration: 40 minutes

URLs and Websites

Learning Objectives

- Define URL and identify its parts
- Distinguish between websites and web pages
- Navigate using URLs in a browser

Timing Breakdown

0-5 min	**Engage**	Ask students: 'Have you ever seen a computer? What did it do?' Write all respons...
5-15 min	**Explore**	Show the parts of a computer on a diagram or real machine. Point to each part an...
15-25 min	**Explain**	Define computer: 'An electronic device that accepts data, processes it, and prod...
25-35 min	**Elaborate**	Present scenario: 'Your school needs to maintain records for 200 students — mark...
35-40 min	**Evaluate**	Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does ...

Engage (5-7 minutes)

Teacher Activity

Ask students: ‘Have you ever seen a computer? What did it do?’ Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: ‘All these involve processing information — that is what computers do.’

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: ‘What do you think this does?’ Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: ‘An electronic device that accepts data, processes it, and produces output.’ Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: ‘Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?’ Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: ‘1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.’ Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 KB, MB, GB — place value and multiplication
Science: Electricity powers all computer components — circuits and conductors
Language: Labeling diagrams builds technical vocabulary and precise writing

Digital Citizen Reminder: Handle computer equipment carefully — it is expensive and shared by all students.

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does. Use pencil and paper only.

Resources Needed:
- Computer diagram or real computer
- labeled chart
- exit ticket paper
- chalkboard

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab — list parts, prices, and total budget
SEND: Touch and identify real computer parts with teacher guidance, simplified labeling

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down a computer system into input, processing, memory, and output components

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.07	Duration: 40 minutes

Introduction to GPS

Learning Objectives

- Define GPS and explain how it works
- Identify devices that use GPS technology
- Describe real-life applications of GPS

Timing Breakdown

0-5 min	Engage	Ask students: 'Have you ever seen a computer? What did it do?' Write all respons...
5-15 min	Explore	Show the parts of a computer on a diagram or real machine. Point to each part an...
15-25 min	Explain	Define computer: 'An electronic device that accepts data, processes it, and prod...
25-35 min	Elaborate	Present scenario: 'Your school needs to maintain records for 200 students — mark...
35-40 min	Evaluate	Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does ...

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: ‘An electronic device that accepts data, processes it, and produces output.’ Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: ‘Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?’ Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: ‘1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.’ Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 KB, MB, GB — place value and multiplication
Science: Electricity powers all computer components — circuits and conductors
Language: Labeling diagrams builds technical vocabulary and precise writing

Digital Citizen Reminder: Handle computer equipment carefully — it is expensive and shared by all students.

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does. Use pencil and paper only.

Resources Needed:
- Computer diagram or real computer
- labeled chart
- exit ticket paper
- chalkboard

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab — list parts, prices, and total budget
SEND: Touch and identify real computer parts with teacher guidance, simplified labeling

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down a computer system into input, processing, memory, and output components

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.08	Duration: 40 minutes

Google Earth

Learning Objectives

- Open Google Earth and navigate the interface
- Search for locations and explore them virtually
- Measure distances between two points

Timing Breakdown

0-5 min	**Engage**	Ask students: 'Have you ever seen a computer? What did it do?' Write all respons...
5-15 min	**Explore**	Show the parts of a computer on a diagram or real machine. Point to each part an...
15-25 min	**Explain**	Define computer: 'An electronic device that accepts data, processes it, and prod...
25-35 min	**Elaborate**	Present scenario: 'Your school needs to maintain records for 200 students — mark...
35-40 min	**Evaluate**	Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does ...

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: ‘An electronic device that accepts data, processes it, and produces output.’ Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: ‘Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?’ Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: ‘1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.’ Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 KB, MB, GB — place value and multiplication
Science: Electricity powers all computer components — circuits and conductors
Language: Labeling diagrams builds technical vocabulary and precise writing

Digital Citizen Reminder: Handle computer equipment carefully — it is expensive and shared by all students.

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does. Use pencil and paper only.

Resources Needed:

- Computer diagram or real computer
- labeled chart
- exit ticket paper
- chalkboard

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab — list parts, prices, and total budget
SEND: Touch and identify real computer parts with teacher guidance, simplified labeling

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down a computer system into input, processing, memory, and output components

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.09	Duration: 40 minutes

Internet Safety

Learning Objectives

- Identify common online dangers
- Practice safe browsing habits
- Create a strong password

Timing Breakdown

0-5 min	Engage	Ask students: 'Have you ever seen a computer? What did it do?' Write all respons...
5-15 min	**Explore**	Show the parts of a computer on a diagram or real machine. Point to each part an...
15-25 min	**Explain**	Define computer: 'An electronic device that accepts data, processes it, and prod...
25-35 min	**Elaborate**	Present scenario: 'Your school needs to maintain records for 200 students — mark...
35-40 min	**Evaluate**	Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does ...

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: ‘An electronic device that accepts data, processes it, and produces output.’ Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: ‘Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?’ Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: ‘1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.’ Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 KB, MB, GB — place value and multiplication
Science: Electricity powers all computer components — circuits and conductors
Language: Labeling diagrams builds technical vocabulary and precise writing

Digital Citizen Reminder: Handle computer equipment carefully — it is expensive and shared by all students.

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does. Use pencil and paper only.

Resources Needed:

- Computer diagram or real computer
- labeled chart
- exit ticket paper
- chalkboard

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab — list parts, prices, and total budget
SEND: Touch and identify real computer parts with teacher guidance, simplified labeling

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down a computer system into input, processing, memory, and output components

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.10 **Duration: 40 minutes**

Email Basics

Learning Objectives

- Identify the parts of an email address
- Compose and send a simple email
- Explain the purpose of inbox, sent, and draft folders

Timing Breakdown

0-5 min	**Engage**	Ask students: 'Have you ever seen a computer? What did it do?' Write all respons...
5-15 min	**Explore**	Show the parts of a computer on a diagram or real machine. Point to each part an...
15-25 min	**Explain**	Define computer: 'An electronic device that accepts data, processes it, and prod...
25-35 min	**Elaborate**	Present scenario: 'Your school needs to maintain records for 200 students — mark...
35-40 min	**Evaluate**	Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does ...

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 KB, MB, GB — place value and multiplication
Science: Electricity powers all computer components — circuits and conductors
Language: Labeling diagrams builds technical vocabulary and precise writing

Digital Citizen Reminder: Handle computer equipment carefully — it is expensive and shared by all students.

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does. Use pencil and paper only.

Resources Needed:

- Computer diagram or real computer
- labeled chart
- exit ticket paper
- chalkboard

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab — list parts, prices, and total budget
SEND: Touch and identify real computer parts with teacher guidance, simplified labeling

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down a computer system into input, processing, memory, and output components

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.11	Duration: 40 minutes

Downloading Files

Learning Objectives

- Download files from the Internet safely
- Identify file types and their extensions
- Organize downloaded files in folders

Timing Breakdown

0-5 min	Engage	Ask students: 'Have you ever seen a computer? What did it do?' Write all respons...
5-15 min	Explore	Show the parts of a computer on a diagram or real machine. Point to each part an...
15-25 min	Explain	Define computer: 'An electronic device that accepts data, processes it, and prod...
25-35 min	Elaborate	Present scenario: 'Your school needs to maintain records for 200 students — mark...
35-40 min	Evaluate	Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does ...

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: ‘An electronic device that accepts data, processes it, and produces output.’ Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: ‘Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?’ Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: ‘1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.’ Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 KB, MB, GB — place value and multiplication
Science: Electricity powers all computer components — circuits and conductors
Language: Labeling diagrams builds technical vocabulary and precise writing

Digital Citizen Reminder: Handle computer equipment carefully — it is expensive and shared by all students.

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does. Use pencil and paper only.

Resources Needed:
- Computer diagram or real computer
- labeled chart
- exit ticket paper
- chalkboard

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab — list parts, prices, and total budget
SEND: Touch and identify real computer parts with teacher guidance, simplified labeling

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down a computer system into input, processing, memory, and output components

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.12	**Duration: 40 minutes**

Chapter 1 Review

Learning Objectives

- Review key concepts from Chapter 1
- Demonstrate skills learned in previous lessons
- Identify areas needing more practice

Timing Breakdown

0-5 min	**Engage**	Ask students: 'Have you ever seen a computer? What did it do?' Write all respons...
5-15 min	**Explore**	Show the parts of a computer on a diagram or real machine. Point to each part an...
15-25 min	**Explain**	Define computer: 'An electronic device that accepts data, processes it, and prod...
25-35 min	**Elaborate**	Present scenario: 'Your school needs to maintain records for 200 students — mark...
35-40 min	**Evaluate**	Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does ...

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: ‘An electronic device that accepts data, processes it, and produces output.’ Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: ‘Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?’ Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: ‘1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.’ Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 KB, MB, GB — place value and multiplication
Science: Electricity powers all computer components — circuits and conductors
Language: Labeling diagrams builds technical vocabulary and precise writing

Digital Citizen Reminder: Handle computer equipment carefully — it is expensive and shared by all students.

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does. Use pencil and paper only.

Resources Needed:
- Computer diagram or real computer
- labeled chart
- exit ticket paper
- chalkboard

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab — list parts, prices, and total budget
SEND: Touch and identify real computer parts with teacher guidance, simplified labeling

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down a computer system into input, processing, memory, and output components

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.01	Duration: 40 minutes

Introduction to Graphics

Learning Objectives

- Define graphics and identify their types
- Distinguish between raster and vector graphics
- Name software used for creating graphics

Timing Breakdown

0-5 min	**Engage**	Ask: 'How many of you have used the internet? What did you do?' Write responses....
5-15 min	**Explore**	Organize pairs. Give 3 minutes to list every ICT device and internet service the...
15-25 min	**Explain**	Define ICT components: Information data types, Communication sharing methods...
25-35 min	**Elaborate**	Present 5 village scenarios: hospital records, bus timings, farmer weather, teac...
35-40 min	**Evaluate**	Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3....

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: URLs follow patterns — domain names use hierarchical structure like place addresses
Science: Fiber optic cables and electromagnetic waves carry internet signals
Language: Writing clear online messages follows communication conventions

Digital Citizen Reminder: Never share personal information online — name, address, phone number, or school name.

Dormitory Task (Paper-and-Pencil)

Write a list of 5 ways ICT is used in your village. For each, name the device and what it does. No computer needed.

Resources Needed:
- Chart paper

- marker pens
- scenario cards
- exit ticket paper

Differentiation

Approaching: Use picture cards of ICT devices with labels for matching
On Level: Complete the scenario matching activity as described
Advanced: Research how ICT is used in a hospital and present findings
SEND: Paired support with visual aids, simplified matching with 3 items

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — identifying the essential components of ICT information, communication, technology

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.02	**Duration: 40 minutes**

Paint Software Basics

Learning Objectives

- Open Paint and identify its interface components
- Use basic tools like pencil, brush, and eraser
- Save a drawing in Paint

Timing Breakdown

0-5 min	**Engage**	Ask: 'How many of you have used the internet? What did you do?' Write responses....
5-15 min	**Explore**	Organize pairs. Give 3 minutes to list every ICT device and internet service the...
15-25 min	**Explain**	Define ICT components: Information data types, Communication sharing methods...
25-35 min	**Elaborate**	Present 5 village scenarios: hospital records, bus timings, farmer weather, teac...
35-40 min	**Evaluate**	Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3....

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: URLs follow patterns — domain names use hierarchical structure like place addresses
Science: Fiber optic cables and electromagnetic waves carry internet signals
Language: Writing clear online messages follows communication conventions

Digital Citizen Reminder: Never share personal information online — name, address, phone number, or school name.

Dormitory Task (Paper-and-Pencil)

Write a list of 5 ways ICT is used in your village. For each, name the device and what it does. No computer needed.

Resources Needed:

- Chart paper

- marker pens
- scenario cards
- exit ticket paper

Differentiation

Approaching: Use picture cards of ICT devices with labels for matching
On Level: Complete the scenario matching activity as described
Advanced: Research how ICT is used in a hospital and present findings
SEND: Paired support with visual aids, simplified matching with 3 items

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — identifying the essential components of ICT information, communication, technology

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.03	Duration: 40 minutes

Drawing Tools

Learning Objectives

- Use shape tools to create geometric figures
- Apply line tools with different thicknesses
- Combine shapes to create simple compositions

Timing Breakdown

0-5 min	Engage	Ask: 'How many of you have used the internet? What did you do?' Write responses....
5-15 min	**Explore**	Organize pairs. Give 3 minutes to list every ICT device and internet service the...
15-25 min	**Explain**	Define ICT components: Information data types, Communication sharing methods...
25-35 min	**Elaborate**	Present 5 village scenarios: hospital records, bus timings, farmer weather, teac...
35-40 min	**Evaluate**	Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3....

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: URLs follow patterns — domain names use hierarchical structure like place addresses
Science: Fiber optic cables and electromagnetic waves carry internet signals
Language: Writing clear online messages follows communication conventions

Digital Citizen Reminder: Never share personal information online — name, address, phone number, or school name.

Dormitory Task (Paper-and-Pencil)

Write a list of 5 ways ICT is used in your village. For each, name the device and what it does. No computer needed.

Resources Needed:

- Chart paper

- marker pens
- scenario cards
- exit ticket paper

Differentiation

Approaching: Use picture cards of ICT devices with labels for matching
On Level: Complete the scenario matching activity as described
Advanced: Research how ICT is used in a hospital and present findings
SEND: Paired support with visual aids, simplified matching with 3 items

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — identifying the essential components of ICT information, communication, technology

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.04	Duration: 40 minutes

Color Selection

Learning Objectives

- Identify primary and secondary colors
- Use the color palette and color picker
- Apply colors to shapes and backgrounds

Timing Breakdown

0-5 min	Engage	Ask: 'How many of you have used the internet? What did you do?' Write responses....
5-15 min	Explore	Organize pairs. Give 3 minutes to list every ICT device and internet service the...
15-25 min	Explain	Define ICT components: Information data types, Communication sharing methods...
25-35 min	Elaborate	Present 5 village scenarios: hospital records, bus timings, farmer weather, teac...
35-40 min	Evaluate	Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3....

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: URLs follow patterns — domain names use hierarchical structure like place addresses
Science: Fiber optic cables and electromagnetic waves carry internet signals
Language: Writing clear online messages follows communication conventions

Digital Citizen Reminder: Never share personal information online — name, address, phone number, or school name.

Dormitory Task (Paper-and-Pencil)

Write a list of 5 ways ICT is used in your village. For each, name the device and what it does. No computer needed.

Resources Needed:

- Chart paper

- marker pens
- scenario cards
- exit ticket paper

Differentiation

Approaching: Use picture cards of ICT devices with labels for matching
On Level: Complete the scenario matching activity as described
Advanced: Research how ICT is used in a hospital and present findings
SEND: Paired support with visual aids, simplified matching with 3 items

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — identifying the essential components of ICT information, communication, technology

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.05	Duration: 40 minutes

Symmetric Designs

Learning Objectives

- Create mirror images using the select tool
- Design patterns with rotational symmetry
- Apply symmetry to real-world design problems

Timing Breakdown

0-5 min	Engage	Ask: 'How many of you have used the internet? What did you do?' Write responses....
5-15 min	**Explore**	Organize pairs. Give 3 minutes to list every ICT device and internet service the...
15-25 min	**Explain**	Define ICT components: Information data types, Communication sharing methods...
25-35 min	**Elaborate**	Present 5 village scenarios: hospital records, bus timings, farmer weather, teac...
35-40 min	**Evaluate**	Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3....

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: URLs follow patterns — domain names use hierarchical structure like place addresses
Science: Fiber optic cables and electromagnetic waves carry internet signals
Language: Writing clear online messages follows communication conventions

Digital Citizen Reminder: Never share personal information online — name, address, phone number, or school name.

Dormitory Task (Paper-and-Pencil)

Write a list of 5 ways ICT is used in your village. For each, name the device and what it does. No computer needed.

Resources Needed:
- Chart paper

- marker pens
- scenario cards
- exit ticket paper

Differentiation

Approaching: Use picture cards of ICT devices with labels for matching
On Level: Complete the scenario matching activity as described
Advanced: Research how ICT is used in a hospital and present findings
SEND: Paired support with visual aids, simplified matching with 3 items

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — identifying the essential components of ICT information, communication, technology

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.06	**Duration: 40 minutes**

Image Editing

Learning Objectives

- Crop images to remove unwanted areas
- Resize images while maintaining proportions
- Adjust brightness and contrast of images

Timing Breakdown

0-5 min	**Engage**	Ask: 'How many of you have used the internet? What did you do?' Write responses....
5-15 min	**Explore**	Organize pairs. Give 3 minutes to list every ICT device and internet service the...
15-25 min	**Explain**	Define ICT components: Information data types, Communication sharing methods...
25-35 min	**Elaborate**	Present 5 village scenarios: hospital records, bus timings, farmer weather, teac...
35-40 min	**Evaluate**	Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3....

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: ‘Which ICT tools help in each?’ Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?’

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: URLs follow patterns — domain names use hierarchical structure like place addresses
Science: Fiber optic cables and electromagnetic waves carry internet signals
Language: Writing clear online messages follows communication conventions

Digital Citizen Reminder: Never share personal information online — name, address, phone number, or school name.

Dormitory Task (Paper-and-Pencil)

Write a list of 5 ways ICT is used in your village. For each, name the device and what it does. No computer needed.

Resources Needed:
- Chart paper

- marker pens
- scenario cards
- exit ticket paper

Differentiation

Approaching: Use picture cards of ICT devices with labels for matching
On Level: Complete the scenario matching activity as described
Advanced: Research how ICT is used in a hospital and present findings
SEND: Paired support with visual aids, simplified matching with 3 items

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — identifying the essential components of ICT information, communication, technology

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.07	Duration: 40 minutes

Layers Concept

Learning Objectives

- Understand what layers are in graphics software
- Create and manage multiple layers
- Arrange layer order to control visibility

Timing Breakdown

0-5 min	Engage	Ask: 'How many of you have used the internet? What did you do?' Write responses....
5-15 min	Explore	Organize pairs. Give 3 minutes to list every ICT device and internet service the...
15-25 min	Explain	Define ICT components: Information data types, Communication sharing methods...
25-35 min	Elaborate	Present 5 village scenarios: hospital records, bus timings, farmer weather, teac...
35-40 min	Evaluate	Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3....

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: URLs follow patterns — domain names use hierarchical structure like place addresses
Science: Fiber optic cables and electromagnetic waves carry internet signals
Language: Writing clear online messages follows communication conventions

Digital Citizen Reminder: Never share personal information online — name, address, phone number, or school name.

Dormitory Task (Paper-and-Pencil)

Write a list of 5 ways ICT is used in your village. For each, name the device and what it does. No computer needed.

Resources Needed:
- Chart paper

- marker pens
- scenario cards
- exit ticket paper

Differentiation

Approaching: Use picture cards of ICT devices with labels for matching
On Level: Complete the scenario matching activity as described
Advanced: Research how ICT is used in a hospital and present findings
SEND: Paired support with visual aids, simplified matching with 3 items

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — identifying the essential components of ICT information, communication, technology

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.08	Duration: 40 minutes

Filters and Effects

Learning Objectives

- Apply basic filters like blur and sharpen
- Use artistic effects to transform images
- Compare original and filtered versions

Timing Breakdown

0-5 min	**Engage**	Ask: 'How many of you have used the internet? What did you do?' Write responses....
5-15 min	**Explore**	Organize pairs. Give 3 minutes to list every ICT device and internet service the...
15-25 min	**Explain**	Define ICT components: Information data types, Communication sharing methods...
25-35 min	**Elaborate**	Present 5 village scenarios: hospital records, bus timings, farmer weather, teac...
35-40 min	**Evaluate**	Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3....

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: URLs follow patterns — domain names use hierarchical structure like place addresses
Science: Fiber optic cables and electromagnetic waves carry internet signals
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Dormitory Task (Paper-and-Pencil)

Write a list of 5 ways ICT is used in your village. For each, name the device and what it does. No computer needed.

Resources Needed:
- Chart paper

- marker pens
- scenario cards
- exit ticket paper

Differentiation

Approaching: Use picture cards of ICT devices with labels for matching
On Level: Complete the scenario matching activity as described
Advanced: Research how ICT is used in a hospital and present findings
SEND: Paired support with visual aids, simplified matching with 3 items

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — identifying the essential components of ICT information, communication, technology

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.09	Duration: 40 minutes

Text in Graphics

Learning Objectives

- Add text to graphics using text tools
- Format text with fonts, sizes, and colors
- Create simple banners and posters with text

Timing Breakdown

0-5 min	Engage	Ask: 'How many of you have used the internet? What did you do?' Write responses....
5-15 min	**Explore**	Organize pairs. Give 3 minutes to list every ICT device and internet service the...
15-25 min	**Explain**	Define ICT components: Information data types, Communication sharing methods...
25-35 min	**Elaborate**	Present 5 village scenarios: hospital records, bus timings, farmer weather, teac...
35-40 min	**Evaluate**	Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3....

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: ‘Which ICT tools help in each?’ Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?’

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: URLs follow patterns — domain names use hierarchical structure like place addresses
Science: Fiber optic cables and electromagnetic waves carry internet signals
Language: Writing clear online messages follows communication conventions

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Dormitory Task (Paper-and-Pencil)

Write a list of 5 ways ICT is used in your village. For each, name the device and what it does. No computer needed.

Resources Needed:
- Chart paper

- marker pens
- scenario cards
- exit ticket paper

Differentiation

Approaching: Use picture cards of ICT devices with labels for matching
On Level: Complete the scenario matching activity as described
Advanced: Research how ICT is used in a hospital and present findings
SEND: Paired support with visual aids, simplified matching with 3 items

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — identifying the essential components of ICT information, communication, technology

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.10	Duration: 40 minutes

File Formats

Learning Objectives

- Identify common image file formats
- Compare qualities and uses of different formats
- Save files in appropriate formats for specific purposes

Timing Breakdown

0-5 min	Engage	Ask: 'How many of you have used the internet? What did you do?' Write responses....
5-15 min	**Explore**	Organize pairs. Give 3 minutes to list every ICT device and internet service the...
15-25 min	**Explain**	Define ICT components: Information data types, Communication sharing methods...
25-35 min	**Elaborate**	Present 5 village scenarios: hospital records, bus timings, farmer weather, teac...
35-40 min	**Evaluate**	Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3....

Engage (5-7 minutes)

Teacher Activity

Ask: ‘How many of you have used the internet? What did you do?’ Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: ‘Information and Communication Technology — all tools that handle information.’

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: ‘What about at the bus stand? Hospital? Shop?’ Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: URLs follow patterns — domain names use hierarchical structure like place addresses
Science: Fiber optic cables and electromagnetic waves carry internet signals
Language: Writing clear online messages follows communication conventions

Digital Citizen Reminder: Never share personal information online — name, address, phone number, or school name.

Dormitory Task (Paper-and-Pencil)

Write a list of 5 ways ICT is used in your village. For each, name the device and what it does. No computer needed.

Resources Needed:

- Chart paper

- marker pens
- scenario cards
- exit ticket paper

Differentiation

Approaching: Use picture cards of ICT devices with labels for matching
On Level: Complete the scenario matching activity as described
Advanced: Research how ICT is used in a hospital and present findings
SEND: Paired support with visual aids, simplified matching with 3 items

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — identifying the essential components of ICT information, communication, technology

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.11	Duration: 40 minutes

Project - School Logo

Learning Objectives

- Apply all learned graphics skills to create a logo
- Plan and execute a design project step by step
- Present and explain design choices to others

Timing Breakdown

0-5 min	Engage	Ask: 'How many of you have used the internet? What did you do?' Write responses....
5-15 min	Explore	Organize pairs. Give 3 minutes to list every ICT device and internet service the...
15-25 min	Explain	Define ICT components: Information data types, Communication sharing methods...
25-35 min	Elaborate	Present 5 village scenarios: hospital records, bus timings, farmer weather, teac...
35-40 min	Evaluate	Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3....

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: URLs follow patterns — domain names use hierarchical structure like place addresses
Science: Fiber optic cables and electromagnetic waves carry internet signals
Language: Writing clear online messages follows communication conventions

Digital Citizen Reminder: Never share personal information online — name, address, phone number, or school name.

Dormitory Task (Paper-and-Pencil)

Write a list of 5 ways ICT is used in your village. For each, name the device and what it does. No computer needed.

Resources Needed:
- Chart paper

- marker pens
- scenario cards
- exit ticket paper

Differentiation

Approaching: Use picture cards of ICT devices with labels for matching
On Level: Complete the scenario matching activity as described
Advanced: Research how ICT is used in a hospital and present findings
SEND: Paired support with visual aids, simplified matching with 3 items

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — identifying the essential components of ICT information, communication, technology

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.12 **Duration: 40 minutes**

Chapter 2 Review

Learning Objectives

- Review all graphics concepts from Chapter 2
- Demonstrate integrated skills in a final project
- Reflect on learning and identify growth areas

Timing Breakdown

0-5 min	**Engage**	Ask: 'How many of you have used the internet? What did you do?' Write responses....
5-15 min	**Explore**	Organize pairs. Give 3 minutes to list every ICT device and internet service the...
15-25 min	**Explain**	Define ICT components: Information data types, Communication sharing methods...
25-35 min	**Elaborate**	Present 5 village scenarios: hospital records, bus timings, farmer weather, teac...
35-40 min	**Evaluate**	Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3....

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: URLs follow patterns — domain names use hierarchical structure like place addresses
Science: Fiber optic cables and electromagnetic waves carry internet signals
Language: Writing clear online messages follows communication conventions

Digital Citizen Reminder: Never share personal information online — name, address, phone number, or school name.

Dormitory Task (Paper-and-Pencil)

Write a list of 5 ways ICT is used in your village. For each, name the device and what it does. No computer needed.

Resources Needed:

- Chart paper

- marker pens
- scenario cards
- exit ticket paper

Differentiation

Approaching: Use picture cards of ICT devices with labels for matching
On Level: Complete the scenario matching activity as described
Advanced: Research how ICT is used in a hospital and present findings
SEND: Paired support with visual aids, simplified matching with 3 items

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — identifying the essential components of ICT information, communication, technology

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.01	Duration: 40 minutes

Types of Communication

Learning Objectives

- Define communication and its purpose
- Identify verbal, non-verbal, and digital communication
- Compare traditional and modern communication methods

Timing Breakdown

0-5 min	**Engage**	Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Di...
5-15 min	**Explore**	Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let...
15-25 min	**Explain**	Define graphics: visual representations of information. Types: painting, drawing...
25-35 min	**Elaborate**	Challenge: 'Create a poster for your school using Paint. Include text, shapes, a...
35-40 min	**Evaluate**	Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital...

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: ‘Which looks better? Why?’ Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; image size depends on resolution
Science: Colors in digital images follow light physics — RGB model
Language: Visual storytelling enhances written narratives and presentations

Digital Citizen Reminder: Only use images you have permission to edit — respect others' creative work.

Dormitory Task (Paper-and-Pencil)

Draw a picture on paper using only a pencil. Then draw the same picture on grid paper, labeling each cell's color. This simulates how computers store images as pixels.

Resources Needed:
- Computer with MS Paint
- sample images
- grid paper
- colored pencils
- poster examples

Differentiation

Approaching: Follow step-by-step screenshot guide for basic Paint tools
On Level: Create a digital drawing using at least 3 tools independently
Advanced: Design a school event poster with text, shapes, and gradients
SEND: Pre-made template to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Pattern Recognition — identifying which digital tool creates which effect

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.02	Duration: 40 minutes

Audio Communication

Learning Objectives

- Understand how sound is used for communication
- Identify devices used for audio communication
- Explain the difference between analog and digital audio

Timing Breakdown

0-5 min	**Engage**	Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Di...
5-15 min	**Explore**	Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let...
15-25 min	**Explain**	Define graphics: visual representations of information. Types: painting, drawing...
25-35 min	**Elaborate**	Challenge: 'Create a poster for your school using Paint. Include text, shapes, a...
35-40 min	**Evaluate**	Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital...

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; image size depends on resolution
Science: Colors in digital images follow light physics — RGB model
Language: Visual storytelling enhances written narratives and presentations

Digital Citizen Reminder: Only use images you have permission to edit — respect others' creative work.

Dormitory Task (Paper-and-Pencil)

Draw a picture on paper using only a pencil. Then draw the same picture on grid paper, labeling each cell's color. This simulates how computers store images as pixels.

Resources Needed:
- Computer with MS Paint
- sample images
- grid paper
- colored pencils
- poster examples

Differentiation

Approaching: Follow step-by-step screenshot guide for basic Paint tools
On Level: Create a digital drawing using at least 3 tools independently
Advanced: Design a school event poster with text, shapes, and gradients
SEND: Pre-made template to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Pattern Recognition — identifying which digital tool creates which effect

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.03	Duration: 40 minutes

Video Communication

Learning Objectives

- Understand how video combines audio and visual elements
- Identify devices and platforms for video communication
- Explain the basic components of a video file

Timing Breakdown

0-5 min	Engage	Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Di...
5-15 min	**Explore**	Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let...
15-25 min	**Explain**	Define graphics: visual representations of information. Types: painting, drawing...
25-35 min	**Elaborate**	Challenge: 'Create a poster for your school using Paint. Include text, shapes, a...
35-40 min	**Evaluate**	Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital...

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; image size depends on resolution
Science: Colors in digital images follow light physics — RGB model
Language: Visual storytelling enhances written narratives and presentations

Digital Citizen Reminder: Only use images you have permission to edit — respect others' creative work.

Dormitory Task (Paper-and-Pencil)

Draw a picture on paper using only a pencil. Then draw the same picture on grid paper, labeling each cell's color. This simulates how computers store images as pixels.

Resources Needed:
- Computer with MS Paint
- sample images
- grid paper
- colored pencils
- poster examples

Differentiation

Approaching: Follow step-by-step screenshot guide for basic Paint tools
On Level: Create a digital drawing using at least 3 tools independently
Advanced: Design a school event poster with text, shapes, and gradients
SEND: Pre-made template to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Pattern Recognition — identifying which digital tool creates which effect

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.04	Duration: 40 minutes

Introduction to Audacity

Learning Objectives

- Identify Audacity as a free audio editing tool
- Navigate the Audacity interface and key controls
- Record and play back a short audio clip

Timing Breakdown

0-5 min	**Engage**	Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Di...
5-15 min	**Explore**	Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let...
15-25 min	**Explain**	Define graphics: visual representations of information. Types: painting, drawing...
25-35 min	**Elaborate**	Challenge: 'Create a poster for your school using Paint. Include text, shapes, a...
35-40 min	**Evaluate**	Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital...

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: ‘Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.’ Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.’

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; image size depends on resolution
Science: Colors in digital images follow light physics — RGB model
Language: Visual storytelling enhances written narratives and presentations

Digital Citizen Reminder: Only use images you have permission to edit — respect others' creative work.

Dormitory Task (Paper-and-Pencil)

Draw a picture on paper using only a pencil. Then draw the same picture on grid paper, labeling each cell’s color. This simulates how computers store images as pixels.

Resources Needed:

- Computer with MS Paint
- sample images
- grid paper
- colored pencils
- poster examples

Differentiation

Approaching: Follow step-by-step screenshot guide for basic Paint tools
On Level: Create a digital drawing using at least 3 tools independently
Advanced: Design a school event poster with text, shapes, and gradients
SEND: Pre-made template to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Pattern Recognition — identifying which digital tool creates which effect

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.05	Duration: 40 minutes

Recording Audio

Learning Objectives

- Set up microphone and adjust recording levels
- Record clear audio with proper techniques
- Identify and reduce common recording errors

Timing Breakdown

0-5 min	**Engage**	Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Di...
5-15 min	**Explore**	Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let...
15-25 min	**Explain**	Define graphics: visual representations of information. Types: painting, drawing...
25-35 min	**Elaborate**	Challenge: 'Create a poster for your school using Paint. Include text, shapes, a...
35-40 min	**Evaluate**	Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital...

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; image size depends on resolution
Science: Colors in digital images follow light physics — RGB model
Language: Visual storytelling enhances written narratives and presentations

Digital Citizen Reminder: Only use images you have permission to edit — respect others' creative work.

Dormitory Task (Paper-and-Pencil)

Draw a picture on paper using only a pencil. Then draw the same picture on grid paper, labeling each cell's color. This simulates how computers store images as pixels.

Resources Needed:
- Computer with MS Paint
- sample images
- grid paper
- colored pencils
- poster examples

Differentiation

Approaching: Follow step-by-step screenshot guide for basic Paint tools
On Level: Create a digital drawing using at least 3 tools independently
Advanced: Design a school event poster with text, shapes, and gradients
SEND: Pre-made template to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Pattern Recognition — identifying which digital tool creates which effect

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.06	**Duration: 40 minutes**

Editing Audio

Learning Objectives

- Cut, copy, and paste audio segments in Audacity
- Remove unwanted sections and silence from recordings
- Apply basic trimming to clean up audio

Timing Breakdown

0-5 min	**Engage**	Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Di...
5-15 min	**Explore**	Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let...
15-25 min	**Explain**	Define graphics: visual representations of information. Types: painting, drawing...
25-35 min	**Elaborate**	Challenge: 'Create a poster for your school using Paint. Include text, shapes, a...
35-40 min	**Evaluate**	Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital...

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; image size depends on resolution
Science: Colors in digital images follow light physics — RGB model
Language: Visual storytelling enhances written narratives and presentations

Digital Citizen Reminder: Only use images you have permission to edit — respect others' creative work.

Dormitory Task (Paper-and-Pencil)

Draw a picture on paper using only a pencil. Then draw the same picture on grid paper, labeling each cell's color. This simulates how computers store images as pixels.

Resources Needed:

- Computer with MS Paint
- sample images
- grid paper
- colored pencils
- poster examples

Differentiation

Approaching: Follow step-by-step screenshot guide for basic Paint tools
On Level: Create a digital drawing using at least 3 tools independently
Advanced: Design a school event poster with text, shapes, and gradients
SEND: Pre-made template to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Pattern Recognition — identifying which digital tool creates which effect

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.07	Duration: 40 minutes

Effects and Filters

Learning Objectives

- Apply audio effects like echo, reverb, and amplification
- Use noise reduction to improve audio quality
- Understand when and how to use different filters

Timing Breakdown

0-5 min	**Engage**	Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Di...
5-15 min	**Explore**	Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let...
15-25 min	**Explain**	Define graphics: visual representations of information. Types: painting, drawing...
25-35 min	**Elaborate**	Challenge: 'Create a poster for your school using Paint. Include text, shapes, a...
35-40 min	**Evaluate**	Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital...

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; image size depends on resolution
Science: Colors in digital images follow light physics — RGB model
Language: Visual storytelling enhances written narratives and presentations

Digital Citizen Reminder: Only use images you have permission to edit — respect others' creative work.

Dormitory Task (Paper-and-Pencil)

Draw a picture on paper using only a pencil. Then draw the same picture on grid paper, labeling each cell's color. This simulates how computers store images as pixels.

Resources Needed:

- Computer with MS Paint
- sample images
- grid paper
- colored pencils
- poster examples

Differentiation

Approaching: Follow step-by-step screenshot guide for basic Paint tools
On Level: Create a digital drawing using at least 3 tools independently
Advanced: Design a school event poster with text, shapes, and gradients
SEND: Pre-made template to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Pattern Recognition — identifying which digital tool creates which effect

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.08	**Duration: 40 minutes**

Mixing Audio

Learning Objectives

- Combine multiple audio tracks into one
- Adjust volume levels and balance between tracks
- Use the timeline to align audio elements

Timing Breakdown

0-5 min	**Engage**	Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Di...
5-15 min	**Explore**	Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let...
15-25 min	**Explain**	Define graphics: visual representations of information. Types: painting, drawing...
25-35 min	**Elaborate**	Challenge: 'Create a poster for your school using Paint. Include text, shapes, a...
35-40 min	**Evaluate**	Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital...

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; image size depends on resolution
Science: Colors in digital images follow light physics — RGB model
Language: Visual storytelling enhances written narratives and presentations

Digital Citizen Reminder: Only use images you have permission to edit — respect others' creative work.

Dormitory Task (Paper-and-Pencil)

Draw a picture on paper using only a pencil. Then draw the same picture on grid paper, labeling each cell's color. This simulates how computers store images as pixels.

Resources Needed:

- Computer with MS Paint
- sample images
- grid paper
- colored pencils
- poster examples

Differentiation

Approaching: Follow step-by-step screenshot guide for basic Paint tools
On Level: Create a digital drawing using at least 3 tools independently
Advanced: Design a school event poster with text, shapes, and gradients
SEND: Pre-made template to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Pattern Recognition — identifying which digital tool creates which effect

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.09	Duration: 40 minutes

Exporting Audio

Learning Objectives

- Export edited audio in common formats like MP3 and WAV
- Choose appropriate quality settings for different uses
- Understand file size versus quality trade-offs

Timing Breakdown

0-5 min	Engage	Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Di...
5-15 min	Explore	Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let...
15-25 min	Explain	Define graphics: visual representations of information. Types: painting, drawing...
25-35 min	Elaborate	Challenge: 'Create a poster for your school using Paint. Include text, shapes, a...
35-40 min	Evaluate	Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital...

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; image size depends on resolution
Science: Colors in digital images follow light physics — RGB model
Language: Visual storytelling enhances written narratives and presentations

Digital Citizen Reminder: Only use images you have permission to edit — respect others' creative work.

Dormitory Task (Paper-and-Pencil)

Draw a picture on paper using only a pencil. Then draw the same picture on grid paper, labeling each cell's color. This simulates how computers store images as pixels.

Resources Needed:
- Computer with MS Paint
- sample images
- grid paper
- colored pencils
- poster examples

Differentiation

Approaching: Follow step-by-step screenshot guide for basic Paint tools
On Level: Create a digital drawing using at least 3 tools independently
Advanced: Design a school event poster with text, shapes, and gradients
SEND: Pre-made template to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Pattern Recognition — identifying which digital tool creates which effect

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.10	Duration: 40 minutes

Video Editing Basics

Learning Objectives

- Import video clips into an editing application
- Trim and arrange clips on a timeline
- Add transitions and text overlays to video

Timing Breakdown

0-5 min	**Engage**	Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Di...
5-15 min	**Explore**	Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let...
15-25 min	**Explain**	Define graphics: visual representations of information. Types: painting, drawing...
25-35 min	**Elaborate**	Challenge: 'Create a poster for your school using Paint. Include text, shapes, a...
35-40 min	**Evaluate**	Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital...

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; image size depends on resolution
Science: Colors in digital images follow light physics — RGB model
Language: Visual storytelling enhances written narratives and presentations

Digital Citizen Reminder: Only use images you have permission to edit — respect others' creative work.

Dormitory Task (Paper-and-Pencil)

Draw a picture on paper using only a pencil. Then draw the same picture on grid paper, labeling each cell's color. This simulates how computers store images as pixels.

Resources Needed:

- Computer with MS Paint
- sample images
- grid paper
- colored pencils
- poster examples

Differentiation

Approaching: Follow step-by-step screenshot guide for basic Paint tools
On Level: Create a digital drawing using at least 3 tools independently
Advanced: Design a school event poster with text, shapes, and gradients
SEND: Pre-made template to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Pattern Recognition — identifying which digital tool creates which effect

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.11	**Duration: 40 minutes**

Project - Audio Story

Learning Objectives

- Plan and script a short audio story
- Record, edit, and mix voice narration with sound effects
- Export and present the final audio story

Timing Breakdown

0-5 min	**Engage**	Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Di...
5-15 min	**Explore**	Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let...
15-25 min	**Explain**	Define graphics: visual representations of information. Types: painting, drawing...
25-35 min	**Elaborate**	Challenge: 'Create a poster for your school using Paint. Include text, shapes, a...
35-40 min	**Evaluate**	Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital...

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; image size depends on resolution
Science: Colors in digital images follow light physics — RGB model
Language: Visual storytelling enhances written narratives and presentations

Digital Citizen Reminder: Only use images you have permission to edit — respect others' creative work.

Dormitory Task (Paper-and-Pencil)

Draw a picture on paper using only a pencil. Then draw the same picture on grid paper, labeling each cell's color. This simulates how computers store images as pixels.

Resources Needed:
- Computer with MS Paint
- sample images
- grid paper
- colored pencils
- poster examples

Differentiation

Approaching: Follow step-by-step screenshot guide for basic Paint tools
On Level: Create a digital drawing using at least 3 tools independently
Advanced: Design a school event poster with text, shapes, and gradients
SEND: Pre-made template to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Pattern Recognition — identifying which digital tool creates which effect

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.12	**Duration: 40 minutes**

Chapter 3 Review

Learning Objectives

- Review all concepts from Audio \\\& Visual Communication
- Demonstrate audio recording, editing, and mixing skills
- Apply chapter knowledge to a practical scenario

Timing Breakdown

0-5 min	**Engage**	Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Di...
5-15 min	**Explore**	Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let...
15-25 min	**Explain**	Define graphics: visual representations of information. Types: painting, drawing...
25-35 min	**Elaborate**	Challenge: 'Create a poster for your school using Paint. Include text, shapes, a...
35-40 min	**Evaluate**	Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital...

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: ‘Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.’ Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.’

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; image size depends on resolution
Science: Colors in digital images follow light physics — RGB model
Language: Visual storytelling enhances written narratives and presentations

Digital Citizen Reminder: Only use images you have permission to edit — respect others' creative work.

Dormitory Task (Paper-and-Pencil)

Draw a picture on paper using only a pencil. Then draw the same picture on grid paper, labeling each cell’s color. This simulates how computers store images as pixels.

Resources Needed:

- Computer with MS Paint
- sample images
- grid paper
- colored pencils
- poster examples

Differentiation

Approaching: Follow step-by-step screenshot guide for basic Paint tools
On Level: Create a digital drawing using at least 3 tools independently
Advanced: Design a school event poster with text, shapes, and gradients
SEND: Pre-made template to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Pattern Recognition — identifying which digital tool creates which effect

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.01	Duration: 40 minutes

Introduction to Data

Learning Objectives

- Define data and distinguish between data and information
- Identify different types of data in daily life
- Explain why computers represent data as numbers

Timing Breakdown

0-5 min	Engage	Play a short audio clip song or speech. Ask: 'How was this recorded? How do we...
5-15 min	Explore	Show Audacity interface or describe it. Demonstrate recording a short clip, pl...
15-25 min	Explain	Define audio communication: sharing information through sound. File formats: MP3...
25-35 min	Elaborate	Task: 'Record a 30-second introduction about yourself. Practice speaking clearly...
35-40 min	Evaluate	Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. W...

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Audio file sizes use binary units KB, MB — powers of 2
Science: Sound waves are vibrations that travel through air — frequency and amplitude
Language: Clear speaking and listening skills support all communication

Digital Citizen Reminder: Be careful what you record and share — audio can be shared widely and permanently.

Dormitory Task (Paper-and-Pencil)

Write a 1-minute speech script on 'My Favorite Festival.' Practice reading it aloud 3 times. Time yourself. Write down one word you stumbled on and practice it.

Resources Needed:

- Audacity if available
- microphone
- speakers
- script template
- timer

Differentiation

Approaching: Follow a pre-written script template, practice with teacher guidance
On Level: Write and record a 30-second introduction independently
Advanced: Create a 2-minute podcast segment with introduction, content, and conclusion
SEND: Use picture prompts to build sentences, practice speaking in small groups

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning a recording script with clear sequence of ideas

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.02	Duration: 40 minutes

Number Systems

Learning Objectives

Timing Breakdown

0-5 min	**Engage**	Play a short audio clip song or speech. Ask: 'How was this recorded? How do we...
5-15 min	**Explore**	Show Audacity interface or describe it. Demonstrate recording a short clip, pl...
15-25 min	**Explain**	Define audio communication: sharing information through sound. File formats: MP3...
25-35 min	**Elaborate**	Task: 'Record a 30-second introduction about yourself. Practice speaking clearly...
35-40 min	**Evaluate**	Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. W...

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Audio file sizes use binary units KB, MB — powers of 2
Science: Sound waves are vibrations that travel through air — frequency and amplitude
Language: Clear speaking and listening skills support all communication

Digital Citizen Reminder: Be careful what you record and share — audio can be shared widely and permanently.

Dormitory Task (Paper-and-Pencil)

Write a 1-minute speech script on 'My Favorite Festival.' Practice reading it aloud 3 times. Time yourself. Write down one word you stumbled on and practice it.

Resources Needed:
- Audacity if available
- microphone
- speakers
- script template
- timer

Differentiation

Approaching: Follow a pre-written script template, practice with teacher guidance
On Level: Write and record a 30-second introduction independently
Advanced: Create a 2-minute podcast segment with introduction, content, and conclusion
SEND: Use picture prompts to build sentences, practice speaking in small groups

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning a recording script with clear sequence of ideas

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.03	Duration: 40 minutes

Binary Conversion

Learning Objectives

- Convert decimal numbers to binary
- Convert binary numbers to decimal
- Verify conversions using the doubling method

Timing Breakdown

0-5 min	**Engage**	Play a short audio clip song or speech. Ask: 'How was this recorded? How do we...
5-15 min	**Explore**	Show Audacity interface or describe it. Demonstrate recording a short clip, pl...
15-25 min	**Explain**	Define audio communication: sharing information through sound. File formats: MP3...
25-35 min	**Elaborate**	Task: 'Record a 30-second introduction about yourself. Practice speaking clearly...
35-40 min	**Evaluate**	Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. W...

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Audio file sizes use binary units KB, MB — powers of 2
Science: Sound waves are vibrations that travel through air — frequency and amplitude
Language: Clear speaking and listening skills support all communication

Digital Citizen Reminder: Be careful what you record and share — audio can be shared widely and permanently.

Dormitory Task (Paper-and-Pencil)

Write a 1-minute speech script on 'My Favorite Festival.' Practice reading it aloud 3 times. Time yourself. Write down one word you stumbled on and practice it.

Resources Needed:
- Audacity if available
- microphone
- speakers
- script template
- timer

Differentiation

Approaching: Follow a pre-written script template, practice with teacher guidance
On Level: Write and record a 30-second introduction independently
Advanced: Create a 2-minute podcast segment with introduction, content, and conclusion
SEND: Use picture prompts to build sentences, practice speaking in small groups

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning a recording script with clear sequence of ideas

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.04	Duration: 40 minutes

Text Representation

Learning Objectives

- Understand ASCII encoding for English text
- Explain how Unicode supports multiple languages
- Convert text to binary using character codes

Timing Breakdown

0-5 min	Engage	Play a short audio clip song or speech. Ask: 'How was this recorded? How do we...
5-15 min	**Explore**	Show Audacity interface or describe it. Demonstrate recording a short clip, pl...
15-25 min	**Explain**	Define audio communication: sharing information through sound. File formats: MP3...
25-35 min	**Elaborate**	Task: 'Record a 30-second introduction about yourself. Practice speaking clearly...
35-40 min	**Evaluate**	Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. W...

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Audio file sizes use binary units KB, MB — powers of 2
Science: Sound waves are vibrations that travel through air — frequency and amplitude
Language: Clear speaking and listening skills support all communication

Digital Citizen Reminder: Be careful what you record and share — audio can be shared widely and permanently.

Dormitory Task (Paper-and-Pencil)

Write a 1-minute speech script on 'My Favorite Festival.' Practice reading it aloud 3 times. Time yourself. Write down one word you stumbled on and practice it.

Resources Needed:
- Audacity if available
- microphone
- speakers
- script template
- timer

Differentiation

Approaching: Follow a pre-written script template, practice with teacher guidance
On Level: Write and record a 30-second introduction independently
Advanced: Create a 2-minute podcast segment with introduction, content, and conclusion
SEND: Use picture prompts to build sentences, practice speaking in small groups

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning a recording script with clear sequence of ideas

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.05	**Duration: 40 minutes**

Image Representation

Learning Objectives

- Understand how images are made of pixels
- Explain color depth and resolution
- Calculate storage needed for a simple image

Timing Breakdown

0-5 min	**Engage**	Play a short audio clip song or speech. Ask: 'How was this recorded? How do we...
5-15 min	**Explore**	Show Audacity interface or describe it. Demonstrate recording a short clip, pl...
15-25 min	**Explain**	Define audio communication: sharing information through sound. File formats: MP3...
25-35 min	**Elaborate**	Task: 'Record a 30-second introduction about yourself. Practice speaking clearly...
35-40 min	**Evaluate**	Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. W...

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Audio file sizes use binary units KB, MB — powers of 2
Science: Sound waves are vibrations that travel through air — frequency and amplitude
Language: Clear speaking and listening skills support all communication

Digital Citizen Reminder: Be careful what you record and share — audio can be shared widely and permanently.

Dormitory Task (Paper-and-Pencil)

Write a 1-minute speech script on 'My Favorite Festival.' Practice reading it aloud 3 times. Time yourself. Write down one word you stumbled on and practice it.

Resources Needed:

- Audacity if available
- microphone
- speakers
- script template
- timer

Differentiation

Approaching: Follow a pre-written script template, practice with teacher guidance
On Level: Write and record a 30-second introduction independently
Advanced: Create a 2-minute podcast segment with introduction, content, and conclusion
SEND: Use picture prompts to build sentences, practice speaking in small groups

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning a recording script with clear sequence of ideas

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.06	Duration: 40 minutes

Audio Representation

Learning Objectives

- Understand how sound waves become digital data
- Explain sampling rate and bit depth in audio
- Calculate file size for different audio quality settings

Timing Breakdown

0-5 min	Engage	Play a short audio clip song or speech. Ask: 'How was this recorded? How do we...
5-15 min	**Explore**	Show Audacity interface or describe it. Demonstrate recording a short clip, pl...
15-25 min	**Explain**	Define audio communication: sharing information through sound. File formats: MP3...
25-35 min	**Elaborate**	Task: 'Record a 30-second introduction about yourself. Practice speaking clearly...
35-40 min	**Evaluate**	Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. W...

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Audio file sizes use binary units KB, MB — powers of 2
Science: Sound waves are vibrations that travel through air — frequency and amplitude
Language: Clear speaking and listening skills support all communication

Digital Citizen Reminder: Be careful what you record and share — audio can be shared widely and permanently.

Dormitory Task (Paper-and-Pencil)

Write a 1-minute speech script on 'My Favorite Festival.' Practice reading it aloud 3 times. Time yourself. Write down one word you stumbled on and practice it.

Resources Needed:
- Audacity if available
- microphone
- speakers
- script template
- timer

Differentiation

Approaching: Follow a pre-written script template, practice with teacher guidance
On Level: Write and record a 30-second introduction independently
Advanced: Create a 2-minute podcast segment with introduction, content, and conclusion
SEND: Use picture prompts to build sentences, practice speaking in small groups

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning a recording script with clear sequence of ideas

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.07	Duration: 40 minutes

Data Processing

Learning Objectives

- Understand the input-process-output cycle
- Identify different types of data processing
- Explain how computers manipulate data automatically

Timing Breakdown

0-5 min	**Engage**	Play a short audio clip song or speech. Ask: 'How was this recorded? How do we...
5-15 min	**Explore**	Show Audacity interface or describe it. Demonstrate recording a short clip, pl...
15-25 min	**Explain**	Define audio communication: sharing information through sound. File formats: MP3...
25-35 min	**Elaborate**	Task: 'Record a 30-second introduction about yourself. Practice speaking clearly...
35-40 min	**Evaluate**	Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. W...

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Audio file sizes use binary units KB, MB — powers of 2
Science: Sound waves are vibrations that travel through air — frequency and amplitude
Language: Clear speaking and listening skills support all communication

Digital Citizen Reminder: Be careful what you record and share — audio can be shared widely and permanently.

Dormitory Task (Paper-and-Pencil)

Write a 1-minute speech script on 'My Favorite Festival.' Practice reading it aloud 3 times. Time yourself. Write down one word you stumbled on and practice it.

Resources Needed:
- Audacity if available
- microphone
- speakers
- script template
- timer

Differentiation

Approaching: Follow a pre-written script template, practice with teacher guidance
On Level: Write and record a 30-second introduction independently
Advanced: Create a 2-minute podcast segment with introduction, content, and conclusion
SEND: Use picture prompts to build sentences, practice speaking in small groups

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning a recording script with clear sequence of ideas

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.08	Duration: 40 minutes

Data Storage

Learning Objectives

- Identify different storage devices and their uses
- Compare primary and secondary storage
- Understand units of digital storage from bytes to gigabytes

Timing Breakdown

0-5 min	Engage	Play a short audio clip song or speech. Ask: 'How was this recorded? How do we...
5-15 min	Explore	Show Audacity interface or describe it. Demonstrate recording a short clip, pl...
15-25 min	Explain	Define audio communication: sharing information through sound. File formats: MP3...
25-35 min	Elaborate	Task: 'Record a 30-second introduction about yourself. Practice speaking clearly...
35-40 min	Evaluate	Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. W...

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Audio file sizes use binary units KB, MB — powers of 2
Science: Sound waves are vibrations that travel through air — frequency and amplitude
Language: Clear speaking and listening skills support all communication

Digital Citizen Reminder: Be careful what you record and share — audio can be shared widely and permanently.

Dormitory Task (Paper-and-Pencil)

Write a 1-minute speech script on 'My Favorite Festival.' Practice reading it aloud 3 times. Time yourself. Write down one word you stumbled on and practice it.

Resources Needed:
- Audacity if available
- microphone
- speakers
- script template
- timer

Differentiation

Approaching: Follow a pre-written script template, practice with teacher guidance
On Level: Write and record a 30-second introduction independently
Advanced: Create a 2-minute podcast segment with introduction, content, and conclusion
SEND: Use picture prompts to build sentences, practice speaking in small groups

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning a recording script with clear sequence of ideas

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.09 Duration: 40 minutes

Data Compression

Learning Objectives

- Understand why data compression is needed
- Distinguish between lossless and lossy compression
- Identify compressed file formats in daily use

Timing Breakdown

0-5 min	Engage	Play a short audio clip song or speech. Ask: 'How was this recorded? How do we...
5-15 min	Explore	Show Audacity interface or describe it. Demonstrate recording a short clip, pl...
15-25 min	Explain	Define audio communication: sharing information through sound. File formats: MP3...
25-35 min	Elaborate	Task: 'Record a 30-second introduction about yourself. Practice speaking clearly...
35-40 min	Evaluate	Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. W...

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Audio file sizes use binary units KB, MB — powers of 2
Science: Sound waves are vibrations that travel through air — frequency and amplitude
Language: Clear speaking and listening skills support all communication

Digital Citizen Reminder: Be careful what you record and share — audio can be shared widely and permanently.

Dormitory Task (Paper-and-Pencil)

Write a 1-minute speech script on 'My Favorite Festival.' Practice reading it aloud 3 times. Time yourself. Write down one word you stumbled on and practice it.

Resources Needed:
- Audacity if available
- microphone
- speakers
- script template
- timer

Differentiation

Approaching: Follow a pre-written script template, practice with teacher guidance
On Level: Write and record a 30-second introduction independently
Advanced: Create a 2-minute podcast segment with introduction, content, and conclusion
SEND: Use picture prompts to build sentences, practice speaking in small groups

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning a recording script with clear sequence of ideas

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.10	Duration: 40 minutes

Data Encryption

Learning Objectives

- Understand why data needs protection
- Explain the concept of encryption and decryption
- Identify everyday uses of encryption

Timing Breakdown

0-5 min	Engage	Play a short audio clip song or speech. Ask: 'How was this recorded? How do we...
5-15 min	Explore	Show Audacity interface or describe it. Demonstrate recording a short clip, pl...
15-25 min	Explain	Define audio communication: sharing information through sound. File formats: MP3...
25-35 min	Elaborate	Task: 'Record a 30-second introduction about yourself. Practice speaking clearly...
35-40 min	Evaluate	Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. W...

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Audio file sizes use binary units KB, MB — powers of 2
Science: Sound waves are vibrations that travel through air — frequency and amplitude
Language: Clear speaking and listening skills support all communication

Digital Citizen Reminder: Be careful what you record and share — audio can be shared widely and permanently.

Dormitory Task (Paper-and-Pencil)

Write a 1-minute speech script on 'My Favorite Festival.' Practice reading it aloud 3 times. Time yourself. Write down one word you stumbled on and practice it.

Resources Needed:

- Audacity if available
- microphone
- speakers
- script template
- timer

Differentiation

Approaching: Follow a pre-written script template, practice with teacher guidance
On Level: Write and record a 30-second introduction independently
Advanced: Create a 2-minute podcast segment with introduction, content, and conclusion
SEND: Use picture prompts to build sentences, practice speaking in small groups

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning a recording script with clear sequence of ideas

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.11	Duration: 40 minutes

Project - Data Analysis

Learning Objectives

- Collect and organize data from a survey
- Use a spreadsheet to enter and calculate data
- Create a chart to present findings visually

Timing Breakdown

0-5 min	Engage	Play a short audio clip song or speech. Ask: 'How was this recorded? How do we...
5-15 min	**Explore**	Show Audacity interface or describe it. Demonstrate recording a short clip, pl...
15-25 min	**Explain**	Define audio communication: sharing information through sound. File formats: MP3...
25-35 min	**Elaborate**	Task: 'Record a 30-second introduction about yourself. Practice speaking clearly...
35-40 min	**Evaluate**	Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. W...

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Audio file sizes use binary units KB, MB — powers of 2
Science: Sound waves are vibrations that travel through air — frequency and amplitude
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Digital Citizen Reminder: Be careful what you record and share — audio can be shared widely and permanently.

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Write a 1-minute speech script on 'My Favorite Festival.' Practice reading it aloud 3 times. Time yourself. Write down one word you stumbled on and practice it.

Resources Needed:
- Audacity if available
- microphone
- speakers
- script template
- timer

Differentiation

Approaching: Follow a pre-written script template, practice with teacher guidance
On Level: Write and record a 30-second introduction independently
Advanced: Create a 2-minute podcast segment with introduction, content, and conclusion
SEND: Use picture prompts to build sentences, practice speaking in small groups

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning a recording script with clear sequence of ideas

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.12	**Duration: 40 minutes**

Chapter 4 Review

Learning Objectives

- Review all concepts from Data Representation and Processing
- Demonstrate binary conversion and data representation skills
- Apply chapter knowledge to solve a real-world problem

Timing Breakdown

0-5 min	**Engage**	Play a short audio clip song or speech. Ask: 'How was this recorded? How do we...
5-15 min	**Explore**	Show Audacity interface or describe it. Demonstrate recording a short clip, pl...
15-25 min	**Explain**	Define audio communication: sharing information through sound. File formats: MP3...
25-35 min	**Elaborate**	Task: 'Record a 30-second introduction about yourself. Practice speaking clearly...
35-40 min	**Evaluate**	Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. W...

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Audio file sizes use binary units KB, MB — powers of 2
Science: Sound waves are vibrations that travel through air — frequency and amplitude
Language: Clear speaking and listening skills support all communication

Digital Citizen Reminder: Be careful what you record and share — audio can be shared widely and permanently.

Dormitory Task (Paper-and-Pencil)

Write a 1-minute speech script on 'My Favorite Festival.' Practice reading it aloud 3 times. Time yourself. Write down one word you stumbled on and practice it.

Resources Needed:

- Audacity if available
- microphone
- speakers
- script template
- timer

Differentiation

Approaching: Follow a pre-written script template, practice with teacher guidance
On Level: Write and record a 30-second introduction independently
Advanced: Create a 2-minute podcast segment with introduction, content, and conclusion
SEND: Use picture prompts to build sentences, practice speaking in small groups

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning a recording script with clear sequence of ideas

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.01	**Duration: 40 minutes**

Introduction to Scratch

Learning Objectives

- Understand what Scratch is and its purpose
- Navigate the Scratch interface and identify key areas
- Create a simple project with a sprite and backdrop

Timing Breakdown

0-5 min	**Engage**	Show a handwritten letter and a typed letter. Ask: 'Which looks more professiona...
5-15 min	**Explore**	Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/...
15-25 min	**Explain**	Define word processing: creating, editing, and formatting text documents. Key fe...
25-35 min	**Elaborate**	Task 1: Write a formal letter to the principal requesting a computer lab. Task 2...
35-40 min	**Evaluate**	Exit ticket: '1. Name two features of a word processor. 2. What is the differenc...

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Spreadsheets apply arithmetic operations — SUM, AVERAGE use algebraic logic
Science: Data tables in science reports use formatting for clarity
Language: Formal letter writing follows conventions — salutation, body, closing

Digital Citizen Reminder: Save your work frequently and use clear file names — losing work is frustrating.

Dormitory Task (Paper-and-Pencil)

Write a formal letter on paper to your teacher requesting something for the school. Use proper format: date, salutation, body, closing. Then draw a table with 5 rows and 3 columns, fill in student names and marks, and calculate totals by hand.

Resources Needed:

- Computer with LibreOffice Writer and Calc
- sample documents
- letter template
- practice data

Differentiation

Approaching: Use a letter template with blanks to fill in
On Level: Write the letter and spreadsheet independently as described
Advanced: Add a table with formatted data to the letter, create a chart in Calc
SEND: Guided letter template, simplified spreadsheet with pre-entered data

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking document creation into steps: content, formatting, saving

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.02	Duration: 40 minutes

Motion Blocks

Learning Objectives

- Identify and use basic Motion blocks in Scratch
- Make a sprite move in different directions
- Control sprite movement using steps and turns

Timing Breakdown

0-5 min	**Engage**	Show a handwritten letter and a typed letter. Ask: 'Which looks more professiona...
5-15 min	**Explore**	Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/...
15-25 min	**Explain**	Define word processing: creating, editing, and formatting text documents. Key fe...
25-35 min	**Elaborate**	Task 1: Write a formal letter to the principal requesting a computer lab. Task 2...
35-40 min	**Evaluate**	Exit ticket: '1. Name two features of a word processor. 2. What is the differenc...

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Spreadsheets apply arithmetic operations — SUM, AVERAGE use algebraic logic
Science: Data tables in science reports use formatting for clarity
Language: Formal letter writing follows conventions — salutation, body, closing

Digital Citizen Reminder: Save your work frequently and use clear file names — losing work is frustrating.

Dormitory Task (Paper-and-Pencil)

Write a formal letter on paper to your teacher requesting something for the school. Use proper format: date, salutation, body, closing. Then draw a table with 5 rows and 3 columns, fill in student names and marks, and calculate totals by hand.

Resources Needed:

- Computer with LibreOffice Writer and Calc
- sample documents
- letter template
- practice data

Differentiation

Approaching: Use a letter template with blanks to fill in
On Level: Write the letter and spreadsheet independently as described
Advanced: Add a table with formatted data to the letter, create a chart in Calc
SEND: Guided letter template, simplified spreadsheet with pre-entered data

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking document creation into steps: content, formatting, saving

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.03	Duration: 40 minutes

Looks Blocks

Learning Objectives

- Use Looks blocks to change sprite appearance
- Switch costumes and backdrops programmatically
- Apply size and visibility effects to sprites

Timing Breakdown

0-5 min	**Engage**	Show a handwritten letter and a typed letter. Ask: 'Which looks more professiona...
5-15 min	**Explore**	Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/...
15-25 min	**Explain**	Define word processing: creating, editing, and formatting text documents. Key fe...
25-35 min	**Elaborate**	Task 1: Write a formal letter to the principal requesting a computer lab. Task 2...
35-40 min	**Evaluate**	Exit ticket: '1. Name two features of a word processor. 2. What is the differenc...

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Spreadsheets apply arithmetic operations — SUM, AVERAGE use algebraic logic
Science: Data tables in science reports use formatting for clarity
Language: Formal letter writing follows conventions — salutation, body, closing

Digital Citizen Reminder: Save your work frequently and use clear file names — losing work is frustrating.

Dormitory Task (Paper-and-Pencil)

Write a formal letter on paper to your teacher requesting something for the school. Use proper format: date, salutation, body, closing. Then draw a table with 5 rows and 3 columns, fill in student names and marks, and calculate totals by hand.

Resources Needed:
- Computer with LibreOffice Writer and Calc
- sample documents
- letter template
- practice data

Differentiation

Approaching: Use a letter template with blanks to fill in
On Level: Write the letter and spreadsheet independently as described
Advanced: Add a table with formatted data to the letter, create a chart in Calc
SEND: Guided letter template, simplified spreadsheet with pre-entered data

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking document creation into steps: content, formatting, saving

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.04	Duration: 40 minutes

Sound Blocks

Learning Objectives

- Add sound effects and music to Scratch projects
- Use Sound blocks to play, stop, and control volume
- Record and use custom sounds in projects

Timing Breakdown

0-5 min	Engage	Show a handwritten letter and a typed letter. Ask: 'Which looks more professiona...
5-15 min	**Explore**	Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/...
15-25 min	**Explain**	Define word processing: creating, editing, and formatting text documents. Key fe...
25-35 min	**Elaborate**	Task 1: Write a formal letter to the principal requesting a computer lab. Task 2...
35-40 min	**Evaluate**	Exit ticket: '1. Name two features of a word processor. 2. What is the differenc...

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: ‘What if you need to calculate marks for 40 students?’ Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students’ marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.’

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Spreadsheets apply arithmetic operations — SUM, AVERAGE use algebraic logic
Science: Data tables in science reports use formatting for clarity
Language: Formal letter writing follows conventions — salutation, body, closing

Digital Citizen Reminder: Save your work frequently and use clear file names — losing work is frustrating.

Dormitory Task (Paper-and-Pencil)

Write a formal letter on paper to your teacher requesting something for the school. Use proper format: date, salutation, body, closing. Then draw a table with 5 rows and 3 columns, fill in student names and marks, and calculate totals by hand.

Resources Needed:
- Computer with LibreOffice Writer and Calc
- sample documents
- letter template
- practice data

Differentiation

Approaching: Use a letter template with blanks to fill in
On Level: Write the letter and spreadsheet independently as described
Advanced: Add a table with formatted data to the letter, create a chart in Calc
SEND: Guided letter template, simplified spreadsheet with pre-entered data

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking document creation into steps: content, formatting, saving

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.05	Duration: 40 minutes

Events and Control

Learning Objectives

- Use Event blocks to start scripts
- Apply Control blocks like wait and repeat
- Understand the role of the green flag and key presses

Timing Breakdown

0-5 min	**Engage**	Show a handwritten letter and a typed letter. Ask: 'Which looks more professiona...
5-15 min	**Explore**	Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/...
15-25 min	**Explain**	Define word processing: creating, editing, and formatting text documents. Key fe...
25-35 min	**Elaborate**	Task 1: Write a formal letter to the principal requesting a computer lab. Task 2...
35-40 min	**Evaluate**	Exit ticket: '1. Name two features of a word processor. 2. What is the differenc...

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Spreadsheets apply arithmetic operations — SUM, AVERAGE use algebraic logic
Science: Data tables in science reports use formatting for clarity
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Resources Needed:
- Computer with LibreOffice Writer and Calc
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- letter template
- practice data

Differentiation

Approaching: Use a letter template with blanks to fill in
On Level: Write the letter and spreadsheet independently as described
Advanced: Add a table with formatted data to the letter, create a chart in Calc
SEND: Guided letter template, simplified spreadsheet with pre-entered data

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking document creation into steps: content, formatting, saving

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.06	Duration: 40 minutes

Variables

Learning Objectives

- Create and use variables in Scratch
- Understand variables as containers for storing values
- Use variables to track scores and counts

Timing Breakdown

0-5 min	Engage	Show a handwritten letter and a typed letter. Ask: 'Which looks more professiona...
5-15 min	Explore	Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/...
15-25 min	Explain	Define word processing: creating, editing, and formatting text documents. Key fe...
25-35 min	Elaborate	Task 1: Write a formal letter to the principal requesting a computer lab. Task 2...
35-40 min	Evaluate	Exit ticket: '1. Name two features of a word processor. 2. What is the differenc...

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Spreadsheets apply arithmetic operations — SUM, AVERAGE use algebraic logic
Science: Data tables in science reports use formatting for clarity
Language: Formal letter writing follows conventions — salutation, body, closing

Digital Citizen Reminder: Save your work frequently and use clear file names — losing work is frustrating.

Dormitory Task (Paper-and-Pencil)

Write a formal letter on paper to your teacher requesting something for the school. Use proper format: date, salutation, body, closing. Then draw a table with 5 rows and 3 columns, fill in student names and marks, and calculate totals by hand.

Resources Needed:

- Computer with LibreOffice Writer and Calc
- sample documents
- letter template
- practice data

Differentiation

Approaching: Use a letter template with blanks to fill in
On Level: Write the letter and spreadsheet independently as described
Advanced: Add a table with formatted data to the letter, create a chart in Calc
SEND: Guided letter template, simplified spreadsheet with pre-entered data

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking document creation into steps: content, formatting, saving

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.07	Duration: 40 minutes

Conditionals

Learning Objectives

- Understand if-then-else logic
- Use conditional blocks to make decisions in programs
- Combine conditions with sensing and operators

Timing Breakdown

0-5 min	**Engage**	Show a handwritten letter and a typed letter. Ask: 'Which looks more professiona...
5-15 min	**Explore**	Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/...
15-25 min	**Explain**	Define word processing: creating, editing, and formatting text documents. Key fe...
25-35 min	**Elaborate**	Task 1: Write a formal letter to the principal requesting a computer lab. Task 2...
35-40 min	**Evaluate**	Exit ticket: '1. Name two features of a word processor. 2. What is the differenc...

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Spreadsheets apply arithmetic operations — SUM, AVERAGE use algebraic logic
Science: Data tables in science reports use formatting for clarity
Language: Formal letter writing follows conventions — salutation, body, closing

Digital Citizen Reminder: Save your work frequently and use clear file names — losing work is frustrating.

Dormitory Task (Paper-and-Pencil)

Write a formal letter on paper to your teacher requesting something for the school. Use proper format: date, salutation, body, closing. Then draw a table with 5 rows and 3 columns, fill in student names and marks, and calculate totals by hand.

Resources Needed:
- Computer with LibreOffice Writer and Calc
- sample documents
- letter template
- practice data

Differentiation

Approaching: Use a letter template with blanks to fill in
On Level: Write the letter and spreadsheet independently as described
Advanced: Add a table with formatted data to the letter, create a chart in Calc
SEND: Guided letter template, simplified spreadsheet with pre-entered data

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking document creation into steps: content, formatting, saving

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.08	Duration: 40 minutes

Loops

Learning Objectives

- Use repeat and forever loops
- Understand when to use count-controlled vs event-controlled loops
- Combine loops with other blocks for complex actions

Timing Breakdown

0-5 min	**Engage**	Show a handwritten letter and a typed letter. Ask: 'Which looks more professiona...
5-15 min	**Explore**	Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/...
15-25 min	**Explain**	Define word processing: creating, editing, and formatting text documents. Key fe...
25-35 min	**Elaborate**	Task 1: Write a formal letter to the principal requesting a computer lab. Task 2...
35-40 min	**Evaluate**	Exit ticket: '1. Name two features of a word processor. 2. What is the differenc...

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Spreadsheets apply arithmetic operations — SUM, AVERAGE use algebraic logic
Science: Data tables in science reports use formatting for clarity
Language: Formal letter writing follows conventions — salutation, body, closing

Digital Citizen Reminder: Save your work frequently and use clear file names — losing work is frustrating.

Dormitory Task (Paper-and-Pencil)

Write a formal letter on paper to your teacher requesting something for the school. Use proper format: date, salutation, body, closing. Then draw a table with 5 rows and 3 columns, fill in student names and marks, and calculate totals by hand.

Resources Needed:
- Computer with LibreOffice Writer and Calc
- sample documents
- letter template
- practice data

Differentiation

Approaching: Use a letter template with blanks to fill in
On Level: Write the letter and spreadsheet independently as described
Advanced: Add a table with formatted data to the letter, create a chart in Calc
SEND: Guided letter template, simplified spreadsheet with pre-entered data

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking document creation into steps: content, formatting, saving

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.09	Duration: 40 minutes

Sensing

Learning Objectives

- Use Sensing blocks to detect interactions
- Detect mouse position, key presses, and sprite collisions
- Create interactive projects using sensing

Timing Breakdown

0-5 min	Engage	Show a handwritten letter and a typed letter. Ask: 'Which looks more professiona...
5-15 min	Explore	Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/...
15-25 min	Explain	Define word processing: creating, editing, and formatting text documents. Key fe...
25-35 min	Elaborate	Task 1: Write a formal letter to the principal requesting a computer lab. Task 2...
35-40 min	Evaluate	Exit ticket: '1. Name two features of a word processor. 2. What is the differenc...

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Spreadsheets apply arithmetic operations — SUM, AVERAGE use algebraic logic
Science: Data tables in science reports use formatting for clarity
Language: Formal letter writing follows conventions — salutation, body, closing

Digital Citizen Reminder: Save your work frequently and use clear file names — losing work is frustrating.

Dormitory Task (Paper-and-Pencil)

Write a formal letter on paper to your teacher requesting something for the school. Use proper format: date, salutation, body, closing. Then draw a table with 5 rows and 3 columns, fill in student names and marks, and calculate totals by hand.

Resources Needed:

- Computer with LibreOffice Writer and Calc
- sample documents
- letter template
- practice data

Differentiation

Approaching: Use a letter template with blanks to fill in
On Level: Write the letter and spreadsheet independently as described
Advanced: Add a table with formatted data to the letter, create a chart in Calc
SEND: Guided letter template, simplified spreadsheet with pre-entered data

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking document creation into steps: content, formatting, saving

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.10	Duration: 40 minutes

Operators

Learning Objectives

- Use mathematical operators for calculations
- Apply comparison and logical operators
- Combine operators in expressions and conditions

Timing Breakdown

0-5 min	**Engage**	Show a handwritten letter and a typed letter. Ask: 'Which looks more professiona...
5-15 min	**Explore**	Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/...
15-25 min	**Explain**	Define word processing: creating, editing, and formatting text documents. Key fe...
25-35 min	**Elaborate**	Task 1: Write a formal letter to the principal requesting a computer lab. Task 2...
35-40 min	**Evaluate**	Exit ticket: '1. Name two features of a word processor. 2. What is the differenc...

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Spreadsheets apply arithmetic operations — SUM, AVERAGE use algebraic logic
Science: Data tables in science reports use formatting for clarity
Language: Formal letter writing follows conventions — salutation, body, closing

Digital Citizen Reminder: Save your work frequently and use clear file names — losing work is frustrating.

Dormitory Task (Paper-and-Pencil)

Write a formal letter on paper to your teacher requesting something for the school. Use proper format: date, salutation, body, closing. Then draw a table with 5 rows and 3 columns, fill in student names and marks, and calculate totals by hand.

Resources Needed:
- Computer with LibreOffice Writer and Calc
- sample documents
- letter template
- practice data

Differentiation

Approaching: Use a letter template with blanks to fill in
On Level: Write the letter and spreadsheet independently as described
Advanced: Add a table with formatted data to the letter, create a chart in Calc
SEND: Guided letter template, simplified spreadsheet with pre-entered data

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking document creation into steps: content, formatting, saving

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.11	Duration: 40 minutes

Project - Animation

Learning Objectives

- Apply all Scratch blocks to create an animation
- Plan and design an animation story
- Debug and refine a complete Scratch project

Timing Breakdown

0-5 min	Engage	Show a handwritten letter and a typed letter. Ask: 'Which looks more professiona...
5-15 min	Explore	Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/...
15-25 min	Explain	Define word processing: creating, editing, and formatting text documents. Key fe...
25-35 min	Elaborate	Task 1: Write a formal letter to the principal requesting a computer lab. Task 2...
35-40 min	Evaluate	Exit ticket: '1. Name two features of a word processor. 2. What is the differenc...

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Spreadsheets apply arithmetic operations — SUM, AVERAGE use algebraic logic
Science: Data tables in science reports use formatting for clarity
Language: Formal letter writing follows conventions — salutation, body, closing

Digital Citizen Reminder: Save your work frequently and use clear file names — losing work is frustrating.

Dormitory Task (Paper-and-Pencil)

Write a formal letter on paper to your teacher requesting something for the school. Use proper format: date, salutation, body, closing. Then draw a table with 5 rows and 3 columns, fill in student names and marks, and calculate totals by hand.

Resources Needed:
- Computer with LibreOffice Writer and Calc
- sample documents
- letter template
- practice data

Differentiation

Approaching: Use a letter template with blanks to fill in
On Level: Write the letter and spreadsheet independently as described
Advanced: Add a table with formatted data to the letter, create a chart in Calc
SEND: Guided letter template, simplified spreadsheet with pre-entered data

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking document creation into steps: content, formatting, saving

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.12	Duration: 40 minutes

Chapter 5 Review

Learning Objectives

- Review all Scratch programming concepts
- Demonstrate understanding through a quiz
- Connect programming skills to real-world applications

Timing Breakdown

0-5 min	**Engage**	Show a handwritten letter and a typed letter. Ask: 'Which looks more professiona...
5-15 min	**Explore**	Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/...
15-25 min	**Explain**	Define word processing: creating, editing, and formatting text documents. Key fe...
25-35 min	**Elaborate**	Task 1: Write a formal letter to the principal requesting a computer lab. Task 2...
35-40 min	**Evaluate**	Exit ticket: '1. Name two features of a word processor. 2. What is the differenc...

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Spreadsheets apply arithmetic operations — SUM, AVERAGE use algebraic logic
Science: Data tables in science reports use formatting for clarity
Language: Formal letter writing follows conventions — salutation, body, closing

Digital Citizen Reminder: Save your work frequently and use clear file names — losing work is frustrating.

Dormitory Task (Paper-and-Pencil)

Write a formal letter on paper to your teacher requesting something for the school. Use proper format: date, salutation, body, closing. Then draw a table with 5 rows and 3 columns, fill in student names and marks, and calculate totals by hand.

Resources Needed:
- Computer with LibreOffice Writer and Calc
- sample documents
- letter template
- practice data

Differentiation

Approaching: Use a letter template with blanks to fill in
On Level: Write the letter and spreadsheet independently as described
Advanced: Add a table with formatted data to the letter, create a chart in Calc
SEND: Guided letter template, simplified spreadsheet with pre-entered data

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking document creation into steps: content, formatting, saving

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.01	**Duration: 40 minutes**

What is Software?

Learning Objectives

- Define software and distinguish it from hardware
- Identify different types of software
- Explain how software makes computers useful

Timing Breakdown

0-5 min	**Engage**	Play a simple Scratch animation on the projector. Ask: 'How do you think this wa...
5-15 min	**Explore**	Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move ...
15-25 min	**Explain**	Define programming: writing instructions for a computer. Scratch uses blocks tha...
25-35 min	**Elaborate**	Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degr...
35-40 min	**Evaluate**	Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why...

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Programming uses angles turn degrees, coordinates x, y position, and counting loops
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Clear variable naming and comments follow language conventions

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or data.

Dormitory Task (Paper-and-Pencil)

On paper, draw a flowchart for making tea: start, boil water, add tea leaves, wait, pour, add milk, stir, end. Use boxes for actions and diamonds for decisions. This is how programmers plan before coding.

Resources Needed:
- Computer with Scratch
- flowchart template
- pencil
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to recreate a given animation
On Level: Create a simple animation with movement and events independently
Advanced: Design an interactive game with score tracking and multiple sprites
SEND: Paper-based block sequencing activity, arrange pre-written instructions in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — creating step-by-step instructions that a computer can follow

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.02	**Duration: 40 minutes**

System Software

Learning Objectives

- Identify system software and its types
- Understand the role of the operating system
- Explain how system software manages computer resources

Timing Breakdown

0-5 min	**Engage**	Play a simple Scratch animation on the projector. Ask: 'How do you think this wa...
5-15 min	**Explore**	Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move ...
15-25 min	**Explain**	Define programming: writing instructions for a computer. Scratch uses blocks tha...
25-35 min	**Elaborate**	Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degr...
35-40 min	**Evaluate**	Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why...

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: ‘Make your sprite move in a square.’ Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What is programming? 2. What does the ‘repeat’ block do? 3. Why is the order of blocks important?’

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Programming uses angles turn degrees, coordinates x, y position, and counting loops
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Clear variable naming and comments follow language conventions

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or data.

Dormitory Task (Paper-and-Pencil)

On paper, draw a flowchart for making tea: start, boil water, add tea leaves, wait, pour, add milk, stir, end. Use boxes for actions and diamonds for decisions. This is how programmers plan before coding.

Resources Needed:
- Computer with Scratch
- flowchart template
- pencil
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to recreate a given animation
On Level: Create a simple animation with movement and events independently
Advanced: Design an interactive game with score tracking and multiple sprites
SEND: Paper-based block sequencing activity, arrange pre-written instructions in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — creating step-by-step instructions that a computer can follow

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.03 **Duration: 40 minutes**

Application Software

Learning Objectives

- Identify application software and its categories
- Understand purpose-specific software tools
- Compare application software with system software

Timing Breakdown

0-5 min	**Engage**	Play a simple Scratch animation on the projector. Ask: 'How do you think this wa...
5-15 min	**Explore**	Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move ...
15-25 min	**Explain**	Define programming: writing instructions for a computer. Scratch uses blocks tha...
25-35 min	**Elaborate**	Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degr...
35-40 min	**Evaluate**	Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why...

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Programming uses angles turn degrees, coordinates x, y position, and counting loops
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Clear variable naming and comments follow language conventions

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or data.

Dormitory Task (Paper-and-Pencil)

On paper, draw a flowchart for making tea: start, boil water, add tea leaves, wait, pour, add milk, stir, end. Use boxes for actions and diamonds for decisions. This is how programmers plan before coding.

Resources Needed:

- Computer with Scratch
- flowchart template
- pencil
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to recreate a given animation
On Level: Create a simple animation with movement and events independently
Advanced: Design an interactive game with score tracking and multiple sprites
SEND: Paper-based block sequencing activity, arrange pre-written instructions in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — creating step-by-step instructions that a computer can follow

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.04	Duration: 40 minutes

Word Processing

Learning Objectives

- Open, create, and save documents in a word processor
- Format text with font, size, color, and alignment
- Use basic word processing features for writing

Timing Breakdown

0-5 min	**Engage**	Play a simple Scratch animation on the projector. Ask: 'How do you think this wa...
5-15 min	**Explore**	Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move ...
15-25 min	**Explain**	Define programming: writing instructions for a computer. Scratch uses blocks tha...
25-35 min	**Elaborate**	Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degr...
35-40 min	**Evaluate**	Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why...

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Programming uses angles turn degrees, coordinates x, y position, and counting loops
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
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Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or data.

Dormitory Task (Paper-and-Pencil)

On paper, draw a flowchart for making tea: start, boil water, add tea leaves, wait, pour, add milk, stir, end. Use boxes for actions and diamonds for decisions. This is how programmers plan before coding.

Resources Needed:

- Computer with Scratch
- flowchart template
- pencil
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to recreate a given animation
On Level: Create a simple animation with movement and events independently
Advanced: Design an interactive game with score tracking and multiple sprites
SEND: Paper-based block sequencing activity, arrange pre-written instructions in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — creating step-by-step instructions that a computer can follow

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.05	Duration: 40 minutes

Spreadsheet Basics

Learning Objectives

- Understand the structure of rows, columns, and cells
- Enter and format data in a spreadsheet
- Use basic formulas for calculations

Timing Breakdown

0-5 min	Engage	Play a simple Scratch animation on the projector. Ask: 'How do you think this wa...
5-15 min	**Explore**	Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move ...
15-25 min	**Explain**	Define programming: writing instructions for a computer. Scratch uses blocks tha...
25-35 min	**Elaborate**	Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degr...
35-40 min	**Evaluate**	Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why...

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Programming uses angles turn degrees, coordinates x, y position, and counting loops
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Clear variable naming and comments follow language conventions

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or data.

Dormitory Task (Paper-and-Pencil)

On paper, draw a flowchart for making tea: start, boil water, add tea leaves, wait, pour, add milk, stir, end. Use boxes for actions and diamonds for decisions. This is how programmers plan before coding.

Resources Needed:

- Computer with Scratch
- flowchart template
- pencil
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to recreate a given animation
On Level: Create a simple animation with movement and events independently
Advanced: Design an interactive game with score tracking and multiple sprites
SEND: Paper-based block sequencing activity, arrange pre-written instructions in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — creating step-by-step instructions that a computer can follow

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.06	**Duration: 40 minutes**

Presentation Software

Learning Objectives

- Create slides with text, images, and backgrounds
- Apply transitions and animations to slides
- Design an effective presentation with clear structure

Timing Breakdown

0-5 min	**Engage**	Play a simple Scratch animation on the projector. Ask: 'How do you think this wa...
5-15 min	**Explore**	Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move ...
15-25 min	**Explain**	Define programming: writing instructions for a computer. Scratch uses blocks tha...
25-35 min	**Elaborate**	Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degr...
35-40 min	**Evaluate**	Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why...

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Programming uses angles turn degrees, coordinates x, y position, and counting loops
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Clear variable naming and comments follow language conventions

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or data.

Dormitory Task (Paper-and-Pencil)

On paper, draw a flowchart for making tea: start, boil water, add tea leaves, wait, pour, add milk, stir, end. Use boxes for actions and diamonds for decisions. This is how programmers plan before coding.

Resources Needed:

- Computer with Scratch
- flowchart template
- pencil
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to recreate a given animation
On Level: Create a simple animation with movement and events independently
Advanced: Design an interactive game with score tracking and multiple sprites
SEND: Paper-based block sequencing activity, arrange pre-written instructions in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — creating step-by-step instructions that a computer can follow

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.07	Duration: 40 minutes

Database Introduction

Learning Objectives

- Understand what a database is and its purpose
- Identify fields, records, and tables
- Recognize real-world uses of databases

Timing Breakdown

0-5 min	**Engage**	Play a simple Scratch animation on the projector. Ask: 'How do you think this wa...
5-15 min	**Explore**	Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move ...
15-25 min	**Explain**	Define programming: writing instructions for a computer. Scratch uses blocks tha...
25-35 min	**Elaborate**	Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degr...
35-40 min	**Evaluate**	Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why...

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Programming uses angles turn degrees, coordinates x, y position, and counting loops
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Clear variable naming and comments follow language conventions

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or data.

Dormitory Task (Paper-and-Pencil)

On paper, draw a flowchart for making tea: start, boil water, add tea leaves, wait, pour, add milk, stir, end. Use boxes for actions and diamonds for decisions. This is how programmers plan before coding.

Resources Needed:
- Computer with Scratch
- flowchart template
- pencil
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to recreate a given animation
On Level: Create a simple animation with movement and events independently
Advanced: Design an interactive game with score tracking and multiple sprites
SEND: Paper-based block sequencing activity, arrange pre-written instructions in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — creating step-by-step instructions that a computer can follow

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.08	Duration: 40 minutes

Utility Software

Learning Objectives

- Identify common utility software types
- Understand how utilities maintain computer health
- Use basic utilities like file manager and antivirus

Timing Breakdown

0-5 min	Engage	Play a simple Scratch animation on the projector. Ask: 'How do you think this wa...
5-15 min	Explore	Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move ...
15-25 min	Explain	Define programming: writing instructions for a computer. Scratch uses blocks tha...
25-35 min	Elaborate	Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degr...
35-40 min	Evaluate	Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why...

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: ‘How do you think this was made?’ Explain: ‘A computer only does what you tell it — programming is giving instructions.’ Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a ‘move 10 steps’ block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Programming uses angles turn degrees, coordinates x, y position, and counting loops
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Clear variable naming and comments follow language conventions

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or data.

Dormitory Task (Paper-and-Pencil)

On paper, draw a flowchart for making tea: start, boil water, add tea leaves, wait, pour, add milk, stir, end. Use boxes for actions and diamonds for decisions. This is how programmers plan before coding.

Resources Needed:

- Computer with Scratch
- flowchart template
- pencil
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to recreate a given animation
On Level: Create a simple animation with movement and events independently
Advanced: Design an interactive game with score tracking and multiple sprites
SEND: Paper-based block sequencing activity, arrange pre-written instructions in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — creating step-by-step instructions that a computer can follow

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.09	**Duration: 40 minutes**

Software Installation

Learning Objectives

- Understand how software is installed on a computer

Timing Breakdown

0-5 min	**Engage**	Play a simple Scratch animation on the projector. Ask: 'How do you think this wa...
5-15 min	**Explore**	Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move ...
15-25 min	**Explain**	Define programming: writing instructions for a computer. Scratch uses blocks tha...
25-35 min	**Elaborate**	Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degr...
35-40 min	**Evaluate**	Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why...

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Programming uses angles turn degrees, coordinates x, y position, and counting loops
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
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On paper, draw a flowchart for making tea: start, boil water, add tea leaves, wait, pour, add milk, stir, end. Use boxes for actions and diamonds for decisions. This is how programmers plan before coding.

Resources Needed:

- Computer with Scratch
- flowchart template
- pencil
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to recreate a given animation
On Level: Create a simple animation with movement and events independently
Advanced: Design an interactive game with score tracking and multiple sprites
SEND: Paper-based block sequencing activity, arrange pre-written instructions in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — creating step-by-step instructions that a computer can follow

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.10	**Duration: 40 minutes**

Software Licenses

Learning Objectives

- Understand what a software license is
- Differentiate between free and paid software
- Recognize the importance of respecting software licenses

Timing Breakdown

0-5 min	**Engage**	Play a simple Scratch animation on the projector. Ask: 'How do you think this wa...
5-15 min	**Explore**	Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move ...
15-25 min	**Explain**	Define programming: writing instructions for a computer. Scratch uses blocks tha...
25-35 min	**Elaborate**	Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degr...
35-40 min	**Evaluate**	Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why...

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

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Dormitory Task (Paper-and-Pencil)

On paper, draw a flowchart for making tea: start, boil water, add tea leaves, wait, pour, add milk, stir, end. Use boxes for actions and diamonds for decisions. This is how programmers plan before coding.

Resources Needed:
- Computer with Scratch
- flowchart template
- pencil
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to recreate a given animation
On Level: Create a simple animation with movement and events independently
Advanced: Design an interactive game with score tracking and multiple sprites
SEND: Paper-based block sequencing activity, arrange pre-written instructions in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — creating step-by-step instructions that a computer can follow

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.11	Duration: 40 minutes

Project - Newsletter

Learning Objectives

- Apply word processing skills to create a newsletter
- Use formatting, images, and layout effectively
- Collaborate and present the final newsletter

Timing Breakdown

0-5 min	Engage	Play a simple Scratch animation on the projector. Ask: 'How do you think this wa...
5-15 min	**Explore**	Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move ...
15-25 min	**Explain**	Define programming: writing instructions for a computer. Scratch uses blocks tha...
25-35 min	**Elaborate**	Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degr...
35-40 min	**Evaluate**	Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why...

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Programming uses angles turn degrees, coordinates x, y position, and counting loops
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Clear variable naming and comments follow language conventions

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or data.

Dormitory Task (Paper-and-Pencil)

On paper, draw a flowchart for making tea: start, boil water, add tea leaves, wait, pour, add milk, stir, end. Use boxes for actions and diamonds for decisions. This is how programmers plan before coding.

Resources Needed:
- Computer with Scratch
- flowchart template
- pencil
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to recreate a given animation
On Level: Create a simple animation with movement and events independently
Advanced: Design an interactive game with score tracking and multiple sprites
SEND: Paper-based block sequencing activity, arrange pre-written instructions in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — creating step-by-step instructions that a computer can follow

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.12	**Duration: 40 minutes**

Chapter 6 Review

Learning Objectives

- Review all software concepts from Chapter 6
- Demonstrate understanding through activities
- Connect software knowledge to everyday computer use

Timing Breakdown

0-5 min	**Engage**	Play a simple Scratch animation on the projector. Ask: 'How do you think this wa...
5-15 min	**Explore**	Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move ...
15-25 min	**Explain**	Define programming: writing instructions for a computer. Scratch uses blocks tha...
25-35 min	**Elaborate**	Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degr...
35-40 min	**Evaluate**	Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why...

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

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Dormitory Task (Paper-and-Pencil)

On paper, draw a flowchart for making tea: start, boil water, add tea leaves, wait, pour, add milk, stir, end. Use boxes for actions and diamonds for decisions. This is how programmers plan before coding.

Resources Needed:
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Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — creating step-by-step instructions that a computer can follow

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.1 | **Duration: 40 minutes**

What is ICT?

Learning Objectives

- Define ICT and identify its components
- Recognize ICT devices used in daily life
- Understand the importance of ICT in modern society

Regional Metaphor: Grain Sorting as Data Processing

Teacher Activity

Ask students: “How do you sort grains at home? What steps do you follow?”

Guide discussion to connect grain sorting to data processing:
- Unsorted grains = Raw data
- Sorting process = Data processing
- Sorted grains = Information

Student Activity

Share their experiences of helping at home with grain sorting.

Give examples of different types of grains they know.

Materials Needed:
- Real grains (rice, wheat) for demonstration
- Chart paper and markers
- ICT textbook

Understanding ICT

Engage (5-7 minutes)

Teacher Activity

Show pictures of ICT devices: computer, mobile phone, tablet, television.

Ask: “How are these devices similar? How are they different?”

Student Activity

Observe the pictures and discuss in pairs.

Share observations with the class.

Explore (10-12 minutes)

Teacher Activity

Explain ICT = Information and Communication Technology

Show the components of ICT:
- Information: Data, text, images, audio, video
- Communication: Sharing, transmitting, receiving
- Technology: Tools, devices, software

Student Activity

Take notes in their notebooks.

Identify ICT components in their daily life.

Explain (10-12 minutes)

Teacher Activity

Live demonstration: Show how a computer processes information.

Use the grain sorting metaphor:
1. Input: Raw grains (unsorted)
2. Processing: Sorting process
3. Output: Sorted grains
4. Storage: Container to store sorted grains

Student Activity

Follow the demonstration.

Draw a diagram showing input ⟶ processing ⟶ output ⟶ storage.

Elaborate (8-10 minutes)

Teacher Activity

Assign Parson's Problem: Arrange the following steps in correct order:
- "We get sorted grains"
- "We put grains in the machine"
- "Machine sorts the grains"
- "We store sorted grains in container"

Student Activity

Work in pairs to arrange the steps.

Present their answer to the class.

Evaluate (5-8 minutes)

Teacher Activity

Conduct exit ticket: Ask students to write:
1. Full form of ICT
2. Two examples of ICT devices
3. One way ICT is useful in daily life

Student Activity

Complete the exit ticket.

Submit before leaving the class.

Cross-Curricular Connections

Mathematics: Numbers and place value connect to binary representation
Science: Sound waves carry telephone signals
Language: Writing clear instructions mirrors algorithm design

Digital Citizen Reminder: Never share your password with anyone except your parents or teacher.

Dormitory Task (Paper-and-Pencil)

Complete the paper-and-pencil activity related to today's topic. Practice the concept without needing a computer.

Resources Needed:
- Chalkboard
- worksheet
- examples of ICT devices (if available)

Differentiation

Approaching: Use word bank with pictures
On Level: Core activity as described
Advanced: Research an ICT device at home
SEND: Paired support, visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down tasks into smaller parts

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson: